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Arthur Cayley

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[647] On the Quartic Surfaces represented by the Equation, Symmetrical Determinant = 0 (continued).

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 46–52; continued from preceding pages.]

(Page 51.) I, in fact, take the terms $(a, \ldots)$ of the determinant to be homogeneous functions of (x, y, z, w) of the degrees

$$\begin{matrix} 0, & 1, & 1, & 0 \\ 1, & 2, & 2, & 1 \\ 1, & 2, & 2, & 1 \\ 0, & 1, & 1, & 0 \end{matrix}$$

respectively, viz. a, d, l are constants, g, h, m, n linear functions, and b, c, f quadric functions of the coordinates; $\nabla = 0$ still represents a quartic surface; and it appears by a general formula that the number of nodes is $= 8$. But we can easily show this directly; and further, that the 8 nodes are the intersections of three quadric surfaces; or say that the quartic surface is octadic. For denoting as before the first minors by $A, \ldots$, then B, C, F are each of them a quadric function of the coordinates, viz. we have

$$\begin{aligned} B &= d(ac - g^2) - cl^2 - an^2 + 2gln, \\ C &= d(ab - h^2) - am^2 - bl^2 + 2hlm, \\ F &= d(gh - af) + l^2 f + mna - nlh - lmg, \end{aligned}$$

and we have identically

$$BC - F^2 = (ad - l^2)\nabla,$$

so that throwing out the constant factor $ad - l^2$, the equation of the surface is

$$BC - F^2 = 0,$$

and it has 8 nodes, the intersections of the three quadric surfaces $B = 0$, $C = 0$, $F = 0$. By equating to zero any other minor of the determinant ∇, we have a surface passing through the 8 nodes; we have for instance the quartic surface

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix}^2 = 0.$$

Suppose now (and in all that follows) that, the degrees being as already mentioned, we further assume that b, c, f are quadric functions of the form $(x,y)^2$, g, h linear functions of the form (x,y); then since each term of ∇ contains either

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

or one of its first minors, it is clear that the line $(x=0, y=0)$ is a double line on the surface. But in the present case there is not any diminution in the number of the nodes; in fact, writing $x=0$, $y=0$, and therefore b, c, f, g, h each $=0$ (but not $a=0$), the minors B, C, F none of them vanish; that is, the line $x=0$, $y=0$ is not a line on any one of the quadric surfaces, and the quadric surfaces intersect as before in an octad of points.

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(Page 52.) The equation $\nabla=0$ thus represents a quartic surface having a double line, and also 8 nodes forming an octad.

We may without loss of generality write $d=0$; in fact, the determinant is unaltered if we add to the fourth column θ times the first column, and then to the fourth line θ times the first line; the determinant is thus of the original form, but in place of d it has $d+2\theta l+\theta^2 a$, which by properly determining θ can be made $=0$. And then changing the original l, m, n, the equation is

$$\nabla = \begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & 0 \end{vmatrix} = 0.$$

Or, writing for shortness,

$$K = \begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

and denoting the minors hereof by $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h})$, then the equation is

$$\nabla = (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h} \mid l, m, n)^2 = 0,$$

where the degree of K is 4, and the degrees of $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$, $\mathfrak{f}$, $\mathfrak{g}$, $\mathfrak{h}$ are 4, 2, 2, 2, 3, 3 respectively, those of l, m, n being 0, 1, 1 respectively.

The nodes are, as before, the intersections of the quadric surfaces $B=0$, $C=0$, $F=0$, viz. (d being now $=0$) the values are

$$\begin{aligned} -B &= cl^2 - 2gln + an^2, \\ -C &= bl^2 - 2hlm + am^2, \\ F &= fl^2 - glm - hln + amn. \end{aligned}$$

But, according to a previous remark, the nodes lie also on the quartic surface $K^2=0$; viz. this is a set of four planes intersecting in the line $x=0$, $y=0$.

Now, in general, any plane through the line $x=0$, $y=0$ meets the surface in this line twice and in a conic; if the plane is $y=\theta x$, we have

$$(a,b,c,f,g,h \mid x,y)^2 = (a, b'x^2, c'x^2, f'x^2, g'x, h'x),$$

where a, b', c', f', g', h' are functions of θ of the degrees $(0,2,2,2,1,1)$ respectively; and thence also

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}) = (\mathfrak{a}', \mathfrak{b}'x^2, \mathfrak{c}'x^2, \mathfrak{f}'x^2, \mathfrak{g}'x^3, \mathfrak{h}'x^3),$$

where $\mathfrak{a}'$, $\mathfrak{b}'$, $\mathfrak{c}'$, $\mathfrak{f}'$, $\mathfrak{g}'$, $\mathfrak{h}'$ are functions of θ of the degrees 4, 2, 2, 2, 3, 3 respectively; the equation of the surface thus becomes $(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}' \mid lx, m, n)^2 = 0$; viz. this is a quadric equation which, combined with the equation $y - \theta x = 0$, determines the

(Page 53.) conic in question. But for each of the planes $K = 0$, we have $(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}' \mid lx, m, n)^2$, a perfect square, or the conic a two-fold line; we have thus the 8 nodes lying in pairs on four lines, say the four "rays," in the four planes $K = 0$ respectively; each of these rays meets the double line $x = 0$, $y = 0$ in a point; and we have thus on the double line 4 points, which are in fact pinch-points of the surface (as to this presently). It has just been stated that for the plane passing through the nodal line and a ray, the conic is a two-fold line (the ray twice) containing upon it a pair of nodes; more properly, the conic is the point-pair composed of the two nodes.

We can find through the nodes four different plane-pairs; in fact, forming the equation

$$B - 2\lambda F + C\lambda^2 = 0,$$

this is

$$l^2(c + 2\lambda f + \lambda^2 b) - 2l(g + \lambda h)(n + \lambda m) + a(n + \lambda m)^2 = 0;$$

or, as this may also be written,

$$[a(n + \lambda m) - l(g + \lambda h)]^2 + l^2 a(b - 2\lambda f + \lambda^2 c) = 0,$$

where b, c, f and therefore also $b - 2\lambda f + \lambda^2 c$ are of the form $(x, y)^2$; say that we have

$$b - 2\lambda f + \lambda^2 c = (p, q, r \mid x, y)^2,$$

where p, q, r are of course quadric functions of λ; determining λ by the quartic equation $pr - q^2 = 0$, we have $b - 2\lambda f + \lambda^2 c$ a perfect square, $= (\alpha x + \beta y)^2$ suppose; and we have thus the plane-pair

$$[a(n + \lambda m) - l(g + \lambda h)]^2 - l^2 a(\alpha x + \beta y)^2 = 0$$

containing the eight nodes; viz. there are four such plane-pairs. The two planes of a plane-pair intersect in a line called an "axis"; that is, we have four axes each meeting the nodal line; and we have thus also through the nodal line and the four axes respectively four planes, which are "pinch-planes" of the quartic surface (as to this presently).

It has just been seen that the equation $B - 2\lambda F + C\lambda^2 = 0$ (where λ is arbitrary) is expressible in the form

$$[a(n + \lambda m) - l(g + \lambda h)]^2 + l^2 a(p, q, r \mid x, y)^2 = 0,$$

viz. this is the equation of a quadric cone having its vertex on the nodal line at the point $x = 0$, $y = 0$, $an - lg + \lambda(am - lh) = 0$; this is, in fact, a cone touching the surface, as at once appears by writing the equation of the cone in the form

$$\frac{1}{C}\{BC - F^2 + (\lambda C - F)^2\} = 0,$$

that is,

$$\frac{1}{C}[-l^2 a\nabla + (\lambda C - F)^2] = 0;$$

we thus see that, taking for vertex any point whatever on the nodal line, there is a circumscribed quadric cone.

(Page 54.) For each of the above-mentioned four values of λ, the quadric cone breaks up into a plane-pair; each plane of the plane-pair is thus a "trope" or plane touching the surface along a conic; viz. this is the conic passing through the intersection of the plane (or say of an axis) with the nodal line and through four nodes of the surface. We have thus 8 tropes, intersecting in pairs in the four axes (and the intersection of each axis with the nodal line being a pinch-point). Moreover, joining the nodes in pairs, we have four rays, each meeting the nodal line, the plane through it and the nodal line being a pinch-plane; this is illustrated in the figure.

[Figure: schematic showing nodal line, axis, pinch-point and pinch-planes.]

As to the pinch-planes and pinch-points, remark first that a plane through the nodal line is in general a bitangent plane, its two points of contact being the points where the conic in such plane meets the nodal line. When the two points of contact come to coincide, the plane is a pinch-plane; viz. this happens when the plane passes through a ray, the conic being then the ray twice repeated. And secondly, at a point on the nodal line there are in general two tangent planes, viz. these are the tangent planes to the quadric cone belonging to such point; when the two tangent-planes come to coincide the point is a pinch-point, and this happens when the point is the intersection of the nodal line with an axis, for then (the quadric cone breaking up into the two tropes through the axis) the two tangent planes become the plane through the axis taken twice.

Each section through the nodal line is a conic, and the polar of the nodal line in regard to this conic is a point; the locus of this point (for different sections through the nodal line) is a right line which may be called simply the "polar." To prove this, considering the section by the plane $y = \theta x$, we have to find the pole of the line $x = 0$ in regard to the conic

$$(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}' \mid lx, m, n)^2 = 0;$$

(Page 55.) this is $lx : m : n = \mathfrak{a}' : \mathfrak{h}' : \mathfrak{g}'$, viz. if $\mathfrak{g} = g_0 x + g_1 y$, $\mathfrak{h} = h_0 x + h_1 y$, this is

$$lx : m : n = a : g_0 + g_1\theta : h_0 + h_1\theta,$$

or joining hereto the equation $y = \theta x$, we have

$$lx : ly : m : n = a : a\theta : g_0 + g_1\theta : h_0 + h_1\theta,$$

where l, a, g_0, g_1, h_0, h_1 are constants; m, n are linear functions of the coordinates (x, y, z, w). The equations represent, it is clear, a right line which is the polar in question; and they may be written

$$\frac{lx}{a} = \frac{h_1 m - g_1 n}{h_0 g_1 - h_1 g_0}, \quad \frac{ly}{a} = \frac{h_0 m - g_0 n}{h_1 g_0 - h_0 g_1}.$$

When the plane passes through a ray, the conic becomes, as was stated, the point-pair composed of the two nodes in such ray; the harmonic in regard to these two points of the intersection of the ray with the nodal line is thus a point on the polar: that is, the polar meets the ray; and the two nodes are situate harmonically in regard to the intersections of the ray with the nodal line and the polar respectively.

The polar may be arrived at in a different manner, viz. if instead of a plane through the nodal line we consider a point on the nodal line, this is the vertex of a circumscribed quadric cone; and taking the polar plane of the nodal line in regard to this cone, then considering the point as variable, the different polar planes all pass through a line which is the polar in question. And hence, taking for the point the intersection of the nodal line with an axis, it appears that the axis meets the polar; and, moreover, that the two tropes through the axis are harmonics in regard to the planes through the axis, and the polar and nodal line respectively.

Collecting the foregoing results, we have a quartic surface as follows:

We have two lines, a nodal line and a polar; meeting each of these, four lines called "rays" and four other lines called "axes." On each ray, harmonically in regard to its intersections with the nodal line and the polar, two nodes of the surface (in all 8 nodes); through each axis, harmonically in regard to the planes through it and the nodal line and the axis respectively, two tropes of the surface (in all 8 tropes). In each trope (or, what is the same thing, in its conic of contact) are 4 nodes; through each node (or, what is the same thing, touching its tangential quadricone) are 4 tropes; the relation of the nodes and tropes may be thus represented, viz. taking the pairs of nodes to be $1, 2$; $3, 4$; $5, 6$; $7, 8$; and those of tropes to be I, II; III, IV; V, VI; VII, VIII; then we have

(Page 56.)

	I	II	III	IV	V	VI	VII	VIII
1	•				•		•	
2		•				•		•
3			•		•			•
4				•		•	•	
5	•		•					•
6		•		•			•	
7	•			•		•		
8		•	•		•			

viz. reading horizontally or vertically, the dots show the tropes through each node, or the nodes in each trope.

The plane through any ray and the nodal line is a pinch-plane of the surface, its point of contact being the intersection of the ray with the nodal line; and the intersection of each axis with the nodal line is a pinch-point of the surface, the tangent plane being the plane through the axis and the nodal line; the surface has thus 4 pinch-planes and 4 pinch-points.

648.

ALGEBRAICAL THEOREM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), p. 53.]

(Page 57.) I WISH to put on record the following theorem, given by me as a Senate-House Problem, January, 1851.

If $\{\alpha + \beta + \gamma + \cdots\}^p$ denote the expansion of $(\alpha + \beta + \gamma + \cdots)^p$, retaining those terms $N\alpha^a\beta^b\gamma^c\cdots$ only in which

$$b + c + d + \cdots > p - 1, \quad c + d + \cdots > p - 2, \text{ \&c., \&c.,}$$

then

$$\begin{aligned} x^n = (x + a)^n - n\{\alpha\}^1(x + a + \beta)^{n-1} + \tfrac{1}{2}n(n-1)\{\alpha + \beta\}^2(x + a + \beta + \gamma)^{n-2} \\ - \tfrac{1}{6}n(n-1)(n-2)\{\alpha + \beta + \gamma\}^3(x + a + \beta + \gamma + \delta)^{n-3} + \text{\&c.} \end{aligned}$$

The theorem, in a somewhat different and imperfectly stated form, is given, Burg, *Crelle*, t. i. (1826), p. 368, as a generalisation of Abel's theorem,

$$\begin{aligned} (x + a)^n = x^n + na(x + \beta)^{n-1} + \tfrac{1}{2}n(n-1)a(a - 2\beta)(x + 2\beta)^{n-2} \\ + \tfrac{1}{6}n(n-1)(n-2)a(a - 3\beta)^2(x + 3\beta)^{n-3} + \text{\&c.} \end{aligned}$$

649.

ADDITION TO MR GLAISHER'S NOTE ON SYLVESTER'S PAPER, "DEVELOPMENT OF AN IDEA OF EISENSTEIN."

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 83, 84.]

(Page 58.) THE formula (11) [in the Note], under a slightly different form, is demonstrated by me in an addition [263] to Sir J. F. W. Herschel's paper "On the formulae investigated by Dr Brinkley, &c.," *Phil. Trans.*, t. CL., 1860, pp. 321–323. The demonstration is in effect as follows: let u denote a series of the form $1+bx+cx^2+dx^3+\cdots$, and let u^i (where i is positive or negative, integer or fractional) denote the development of the i-th power of u, continued up to the term which involves x^n, the terms involving higher powers of x being rejected; u^0, u^1, u^2, and generally u^s will denote in like manner the developments of these powers up to the terms involving x^n, or, what is the same thing, they will be the values of u^i corresponding to $i = 0, 1, 2, \ldots, s$. By the formula

$$u^i = 1 + \tfrac{i}{1}(u-1) + \tfrac{i \cdot i-1}{1 \cdot 2}(u-1)^2 + \cdots \text{ as far as the term involving } (u-1)^n,$$ u^i is a rational

and integral function of i of the degree n, and can therefore be expressed in terms of the values u^0, u^1, u^2, ..., u^n which correspond to $i = 0, 1, 2, \ldots, n$. Let s have any one of the last-mentioned values, then the expression

$$\frac{i \cdot i-1 \cdot i-2 \cdots i-n}{i-s} \cdot \frac{1}{s \cdot s-1 \cdots 2 \cdot 1 \cdot -1 \cdot -2 \cdots -(n-s)},$$

which as regards i is a rational and integral function of the degree n (the factor $i-s$ which occurs in the numerator and denominator being of course omitted), vanishes for each of the values $i = 0, 1, 2, \ldots, n$, except only for the value $i = s$, in which case it becomes equal to unity. The required formula is thus seen to be

$$u^i = \Sigma \left\{ \frac{i \cdot i-1 \cdot i-2 \cdots i-n}{i-s} \cdot \frac{1}{s \cdot s-1 \cdots 2 \cdot 1 \cdot -1 \cdot -2 \cdots -(n-s)} u^s \right\},$$

(Page 59.) where the summation extends to the several values $s = 0, 1, 2, \ldots, n$; or, what is the same thing, it is

$$u^i = \Sigma \left\{ \frac{i \cdot i-1 \cdot i-2 \cdots i-n}{i-s} \cdot \frac{(-)^{n-s} 1}{1 \cdot 2 \cdots s \cdot 1 \cdot 2 \cdots n-s} u^s \right\},$$

or, changing the sign of i, it is

$$u^{-i} = \Sigma \left\{ \frac{i \cdot i+1 \cdot i+2 \cdots i+n}{i+s} \cdot \frac{(-)^{s} 1}{1 \cdot 2 \cdots s \cdot 1 \cdot 2 \cdots n-s} u^s \right\},$$

where, as before, s has the values $0, 1, 2, \ldots, n$ successively. Or, what is the same thing, we have

$$C_{-i,n} = \Sigma \left\{ \frac{i \cdot i+1 \cdot i+2 \cdots i+n}{i+s} \cdot \frac{(-)^{s} 1}{1 \cdot 2 \cdots s \cdot 1 \cdot 2 \cdots n-s} C_{s,n} \right\},$$

where the term corresponding to $s = 0$, as containing the factor $C_{0,n}$ vanishes except in the case $n = 0$ (for which it is $= 1$); and omitting this evanescent term, this is in fact the formula (11).

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650.

ON A QUARTIC SURFACE WITH TWELVE NODES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 103–106.]

(Page 60.) WRITE for shortness

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, & af &= p, \\ b &= \gamma - \alpha, & g &= \beta - \delta, & bg &= q, \\ c &= \alpha - \beta, & h &= \gamma - \delta, & ch &= r; \end{aligned}$$

then, θ being a variable parameter, the surface in question is the envelope of the quadric surface

$$(\alpha+\theta)^2\, aghX^2 + (\beta+\theta)^2\, bhfY^2 + (\gamma+\theta)^2\, cfgZ^2 + (\delta+\theta)^2\, abcW^2 = 0;$$

viz. this is

$$\Sigma\theta^2 aghX^2 \cdot \Sigma aghX^2 - \Sigma a\, aghX^2 = 0.$$

There are no terms in X^4, &c.; the coefficient of Y^2Z^2 is

$$\gamma^2 cfg \cdot bfh + \beta^2 bfh \cdot cfg - 2\beta\gamma\, bfh \cdot cfg,$$

which is

$$= bcfgh\,(\beta-\gamma)^2, \ = a^2bcf^2gh, \ = abcfgh \cdot p.$$

Hence the whole equation divides by $abcfgh$, and throwing out this factor, the result is

$$p(Y^2Z^2 + X^2W^2) + q(Z^2X^2 + Y^2W^2) + r(X^2Y^2 + Z^2W^2) = 0,$$

or, observing that $p+q+r=0$, this may also be written

$$p(YZ+XW)^2 + q(ZX+YW)^2 + r(XY+ZW)^2 = 0,$$

(Page 61.) and also

$$p(YZ-XW)^2 + q(ZX-YW)^2 + r(XY-ZW)^2 = 0.$$

The more general equation

$$(p,q,r,l,m,n \mid YZ+XW,\ ZX+YW,\ XY+ZW)^2 = 0$$

represents a quartic surface (octadic) having the 8 nodes

$$\begin{matrix} (1,0,0,0), & (\bar{1},1,1,1), \\ (0,1,0,0), & (1,\bar{1},1,1), \\ (0,0,1,0), & (1,1,\bar{1},1), \\ (0,0,0,1), & (1,1,1,\bar{1}). \end{matrix}$$

We have

$d_xU =$	$d_yU =$
$p \cdot XW^2 + YZW$	$p \cdot YZ^2 + XZW$
$q \cdot YW^2 + YZW$	$q \cdot YW^2 + XZW$
$r \cdot ZW^2 + YZW$	$r \cdot YX^2 + XZW$
$l \cdot 2XYZ + W(Y^2+Z^2)$	$l \cdot 2XYW + Z(W^2+X^2)$
$m \cdot 2XYW + Z(W^2+Y^2)$	$m \cdot 2YZX + W(Z^2+X^2)$
$n \cdot 2XZW + Y(W^2+Z^2)$	$n \cdot 2YZW + X(W^2+Z^2)$,

$d_zU =$	$d_wU =$
$p \cdot Y^2Z + XYW$	$p \cdot X^2W + XYZ$
$q \cdot X^2Z + XYW$	$q \cdot Y^2W + XYZ$
$r \cdot W^2Z + XYW$	$r \cdot Z^2W + XYZ$
$l \cdot 2ZXW + Y(W^2 + X^2)$	$l \cdot 2WYZ + X(Y^2 + Z^2)$
$m \cdot 2YZW + X(W^2 + Y^2)$	$m \cdot 2WZX + Y(Z^2 + X^2)$
$n \cdot 2ZXY + W(X^2 + Y^2),$	$n \cdot 2WXY + Z(X^2 + Y^2).$

Hence there will be a node

$$\begin{array}{ll} 1, 1, 1, 1, & \ldots\ p + q + r + 2l + 2m + 2n = 0, \\ \bar{1}, 1, 1, 1, & \ldots\ p + q + r - 2l + 2m - 2n = 0, \\ 1, \bar{1}, 1, 1, & \ldots\ p + q + r + 2l - 2m + 2n = 0, \\ 1, 1, \bar{1}, 1, & \ldots\ p + q + r - 2l - 2m + 2n = 0; \end{array}$$

or say there will be

$$\begin{array}{cl} \text{1 of these nodes} & p + q + r + 2l + 2m + 2n = 0, \\ 2 & p + q + r + 2l = 0,\ m + n = 0, \\ 3 & p + q + r = 2l = -2m = -2n, \\ 4 & p + q + r = 0,\ l = 0,\ m = 0,\ n = 0; \end{array}$$

(Page 62.) viz. the surface having the 12 nodes is the original surface

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2,$$

where

$$p + q + r = 0.$$

The Jacobian of the quadrics

$$YZ + XW = 0, \quad ZX + YW = 0, \quad XY + ZW = 0,$$

is

$$\begin{vmatrix} W, & Z, & Y, & X \\ Z, & W, & X, & Y \\ Y, & X, & W, & Z \end{vmatrix} = 0;$$

viz. the equations are

$$\begin{aligned} X^3 - X(Y^2 + Z^2 + W^2) + 2YZW &= 0, \\ Y^3 - Y(Z^2 + X^2 + W^2) + 2ZXW &= 0, \\ Z^3 - Z(X^2 + Y^2 + W^2) + 2XYW &= 0, \\ W^3 - W(X^2 + Y^2 + Z^2) + 2XYZ &= 0, \end{aligned}$$

each of which is satisfied in virtue of any one of the pairs of equations

$$\begin{array}{ll} (Y - Z = 0,\ X - W = 0) & (Y + Z = 0,\ X + W = 0), \\ (Z - X = 0,\ Y - W = 0) & (Z + X = 0,\ Y + W = 0), \\ (X - Y = 0,\ Z - W = 0) & (X + Y = 0,\ Z + W = 0). \end{array}$$

so that the Jacobian curve is, in fact, the six lines represented by these equations.

Any two of the three tetrads form an octad, the 8 points of intersection of three quadric surfaces: a figure representing the relation of the 12 points to each other may be constructed without difficulty.

Each tetrad is a sibi-conjugate tetrad *quoad* the quadric $X^2 + Y^2 + Z^2 + W^2 = 0$. The three tetrads are not on the same quadric surface.

651.

ON A SPECIAL SURFACE OF MINIMUM AREA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 190–196.]

(Page 63.) A VERY remarkable form of the surface of minimum area was obtained by Prof. Schwarz in his memoir "Bestimmung einer speciellen Minimal-fläche," Berlin, 1871, [*Ges. Werke*, t. I., pp. 6–125], crowned by the Academy of Sciences at Berlin. The equation of the surface is

$$1+\mu\nu+\nu\lambda+\lambda\mu=0,$$

where λ, μ, ν are functions of x, y, z respectively, viz.

$$x=-\int\frac{d\theta}{\sqrt{(\frac{1}{4}\theta^4+\frac{1}{2}\theta^2+\frac{1}{4})}},$$

and y, z are the same functions of μ, ν respectively. A direct verification of the theorem that this is a surface of minimum area, satisfying, that is, the differential equation

$$r(1+q^2)-2pqs+t(1+p^2)=0,$$

is given in the memoir; but the investigation may be conducted in quite a different manner, so as to be at once symmetrical and somewhat more general, viz. we may enquire whether there exists a surface of minimum area

$$1+\mu\nu+\nu\lambda+\lambda\mu=0,$$

where the determining equations are

$$\begin{aligned}\lambda'^2&=a\lambda^4+b\lambda^2+c,\\ \mu'^2&=a\mu^4+b\mu^2+c,\\ \nu'^2&=a\nu^4+b\nu^2+c,\end{aligned}$$

(Page 64.) $\left(\lambda'=\frac{d\lambda}{dx},\ \&c.\right)$. I find that the coefficients a, b, c must satisfy four homogeneous quadric equations, which, in fact, admit of simultaneous solution, and that in three distinct ways; viz. assuming $a=1$, the solutions are

$$\begin{aligned}&a=1,\quad b=\tfrac{1}{2},\quad c=1,\\ &a=1,\quad b=-2,\quad c=1,\\ &a=1,\quad b=-\tfrac{3}{2},\quad c=-\tfrac{1}{2};\end{aligned}$$

that is,

$$\lambda'^2=\lambda^4+\tfrac{1}{2}\lambda^2+1\ \{=\ \tfrac{1}{4}(\tfrac{4}{3}\lambda^4+\tfrac{2}{3}\lambda^2+\tfrac{4}{3})\},$$

which gives Schwarz's surface;

$$\lambda'^2=\lambda^4-2\lambda^2+1 \text{ or } \lambda'=\pm(\lambda^2-1),$$

which, it is easy to see, gives only $x+y+z=\text{const.}$; and

$$\lambda'^2=\lambda^4-\tfrac{3}{2}\lambda^2-\tfrac{1}{2},\ =(\lambda^2-1)(\lambda^2+\tfrac{1}{2}),$$

which is a surface similar in its nature to Schwarz's surface.

The investigation is as follows: the condition to be satisfied by a surface of minimum area $U = 0$ is

$$(\mathfrak{a}+\mathfrak{b}+\mathfrak{c})(X^2+Y^2+Z^2) - (\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{f},\mathfrak{g},\mathfrak{h} \mid X,Y,Z)^2 = 0,$$

where (X,Y,Z) are the first derived coefficients and $(\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{f},\mathfrak{g},\mathfrak{h})$ the second derived coefficients of U in regard to the coordinates. Considering U as a function of λ, μ, ν, which are functions of x, y, z respectively, and writing (L,M,N) and $(\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{f},\mathfrak{g},\mathfrak{h})$ for the first and second derived functions of U in regard to λ, μ, ν, also λ', λ'' for the first and second derived functions of λ in regard to x, and so for μ, μ'' and ν, ν'' we have:

$$(X,Y,Z) = (L\lambda', M\mu', N\nu'),$$

$$(\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{f},\mathfrak{g},\mathfrak{h}) = (\mathsf{a}\lambda'^2 + L\lambda'',\ \mathsf{b}\mu'^2 + M\mu'',\ \mathsf{c}\nu'^2 + N\nu'',\ \mathsf{f}\mu'\nu',\ \mathsf{g}\nu'\lambda',\ \mathsf{h}\lambda'\mu'),$$

and for the particular surface $U = 1+\mu\nu+\nu\lambda+\lambda\mu = 0$, the values are

$$(L,M,N,\mathsf{a},\mathsf{b},\mathsf{c},\mathsf{f},\mathsf{g},\mathsf{h}) = (\mu+\nu,\ \nu+\lambda,\ \lambda+\mu,\ 0,0,0,1,1,1).$$

Hence the condition is found to be

$$\begin{aligned}
&2\mu'^2\nu'^2(\lambda+\mu)(\lambda+\nu)\\
&\quad + 2\nu'^2\lambda'^2(\mu+\nu)(\mu+\lambda)\\
&\quad + 2\lambda'^2\mu'^2(\nu+\lambda)(\nu+\mu)\\
&\quad - \lambda''(\mu+\nu)\{(\lambda+\nu)^2\mu'^2 + (\lambda+\mu)^2\nu'^2\}\\
&\quad - \mu''(\nu+\lambda)\{(\mu+\lambda)^2\nu'^2 + (\mu+\nu)^2\lambda'^2\}\\
&\quad - \nu''(\lambda+\mu)\{(\nu+\mu)^2\lambda'^2 + (\nu+\lambda)^2\mu'^2\} = 0,
\end{aligned}$$

(Page 65.) or say this is

$$\Sigma 2\mu'^2\nu'^2(\lambda+\mu)(\lambda+\nu) - \Sigma\lambda''(\mu+\nu)\{(\lambda+\nu)^2\mu'^2 + (\lambda+\mu)^2\nu'^2\} = 0.$$

We have to write in this equation $\lambda'^2 = a\lambda^4 + b\lambda^2 + c$, and therefore $\lambda'' = 2a\lambda^3 + b\lambda$, &c.; the left-hand side, call it Ω, is a symmetrical function of λ, μ, ν, and is consequently expressible as a rational function of

$$\begin{aligned}
p, &\ = \lambda+\mu+\nu,\\
q, &\ = \mu\nu+\nu\lambda+\lambda\mu,\\
r, &\ = \lambda\mu\nu.
\end{aligned}$$

We ought to have $\Omega = 0$, not identically, but in virtue of the equation $1+q = 0$, that is, Ω should divide by $1+q$; or, what is the same thing, Ω should vanish on writing therein $q = -1$.

To effect the reduction as easily as possible, observe that we have $(\lambda+\mu)(\lambda+\nu) = \lambda^2+q$; and therefore

$$\Sigma\mu'^2\nu'^2(\lambda+\mu)(\lambda+\nu) = \Sigma\lambda^2\mu'^2\nu'^2 + q\Sigma\mu'^2\nu'^2.$$

Similarly, in the second term,

$$(\mu+\nu)(\lambda+\nu)^2 = (\nu+\lambda)(\nu^2+q) \text{ and } (\mu+\nu)(\lambda+\mu)^2 = (\mu+\lambda)(\mu^2+q).$$

The complete value of Ω thus is

$$\Omega = 2(Aq+B) - [(C+D)q+E+F],$$

where

$$A = \Sigma\lambda^2\mu'^2\nu'^2, \qquad B = \Sigma\mu'^2\nu'^2,$$
$$C = \Sigma\lambda\lambda''(\nu\mu'^2 + \mu\nu'^2), \qquad D = \Sigma\lambda''(\nu\mu'^2 + \mu\nu'^2),$$
$$E = \Sigma\lambda\lambda''(\mu'^2 + \nu'^2), \qquad F = \Sigma\lambda''(\nu\mu'^2 + \mu\nu'^2).$$

We find without difficulty

$$\begin{aligned} A = a^2 & (q^4 - 4q^2pr + 2r^2 + 2p^2r^2) \\ & + ab\,(-2q^3 + qp^2 + 4qr^2 - 3r^2 - 2p^2r) \\ & + ac\,(\ 4q^2 - 3q^2p^2 + 8pr + 2p^2) \\ & + b^2\,(\ q^2 - 2pr) \\ & + bc\,(-4q + 2p^2) \\ & + c^2\,(\ 3), \end{aligned}$$

$$\begin{aligned} B = a^2 & (\ q^2r^2 + 2pr^3) \\ & + ab\,(-4qr^2 + 2p^2r^2) \\ & + ac\,(-2q^3 + q^2p^2 + 4qpr - 3r^2 - 2p^2r) \\ & + b^2\,(\ 3r^2) \\ & + bc\,(\ 2q^2 - 4pr) \\ & + c^2\,(-2q + 2p^2), \end{aligned}$$

(Page 66.)

$$\begin{aligned} C = a^2 & (\ 4q^2 - 16q^2pr + 16pr^2 + 8p^2r^2) \\ & + ab\,(-6q^3 + 3q^2p^2 + 12qpr - 9r^2 - 6p^2r) \\ & + ac\,(\ 8q^2 - 16qp^2 + 16pr + 4p^4) \\ & + b^2\,(\ 2q^2 - 4pr) \\ & + bc\,(-4q + 2p^2), \end{aligned}$$

$$\begin{aligned} D = a^2 & (\ 4q^2pr - 2qr^2 - 4p^2r^2) \\ & + ab\,(-4qpr + 2p^2r) \\ & + ac\,(-4q^2 + 2qp^2 - 2pr) \\ & + b^2\,(\ 2pr) \\ & + bc\,(\ 2q), \end{aligned}$$

$$\begin{aligned} E = a^2 & (\ 4q^2r - 4pr^2) \\ & + ab\,(-12qr^2 + 6p^2r^2) \\ & + ac\,(-4q^2 + 2qp^2 + 8qpr - 6r^2 - 4p^2r) \\ & + bc\,(\ 2q - 4pr), \end{aligned}$$

$$\begin{aligned} F = a^2 & (\ 4pr^2) \\ & + ab\,(\ ) \\ & + ac\,(\ 4q^2 - 12qpr + 12p^2r) \\ & + b^2\,(\ qpr - 3r^2) \\ & + bc\,(-2q^2 + qp^2 - pr), \end{aligned}$$

where in each line the terms are arranged according to their order in p, r.

Substituting, we find

$$\begin{aligned}\Omega = a^2\,(& -2q^3+6q^2pr-8qr^2 \quad) \\ &+ab\,(\ 2q^2-q^2pr+qr^2-4r^2) \\ &+ac\,[-2q^3p^2+14qpr-12r^2] \\ &+b^2\,(\ -3qpr+3r^2) \\ &+bc\,(\ -2q^2+qp^2-3pr) \\ &+c^2\,(\ 2q+2p^2);\end{aligned}$$

viz. writing $q=-1$, this is

$$\begin{aligned}\Omega = a^2\,(& \ 2-6pr-8r^2) \\ &+ab\,(\ 2+p^2-pr-4r^2) \\ &+ac\,(\ -2p^2-14pr-12r^2) \\ &+b^2\,(\ 3pr+3r^2) \\ &+bc\,(-2-p^2-3pr) \\ &+c^2\,(-2-2p^2);\end{aligned}$$

(Page 67.) or, what is the same thing, it is

$$\begin{aligned}&(\ 2a^2+2ab-2bc-2c^2) \\ &+p^2(\ ab-2ac-bc+2c^2) \\ &+pr(-6a^2-ab-14ac+3b^2-3bc) \\ &+r^2(-8a^2-4ab-12ac+3b^2)\end{aligned}$$

so that, writing for convenience $a=1$, the equations to be satisfied are

$$\begin{aligned}2-2c^2+2(1-c)b&=0, \\ -2c+2c^2+(1-c)b&=0, \\ -6-14c+3b^2-(1+3c)b&=0, \\ -8-12c+3b^2-4b&=0.\end{aligned}$$

The first and second are $(1-c)(2+2c+2b)=0$ and $(1-c)(-2c+b)=0$; viz. they give $c=1$, or else $b=-\frac{1}{2}$, $c=\frac{1}{2}$. In the former case, the third and fourth equations each become $3b^2-4b-20=0$, that is $(3b-10)(b+2)=0$; in the latter case, they are satisfied identically; hence we have for a, b, c the three systems of values mentioned at the beginning.

This completes the investigation; but it is interesting to find the values assumed by the other factor of Ω on substituting therein for a, b, c the foregoing several systems of values. We have in general

$$\begin{aligned}
\Omega = &- 2a^2q^3 + 2abq^2 - 2bcq^2 + 2c^2q \\
&+ p^2(-abq^2 - 2acq^2 + bcq + 2c^2) \\
&+ pr[6a^2q^2 - abq + (14ac - 3b^2)q - 3bc] \\
&+ r^2[-8a^2q + 4abq - 12ac + 3b^2] \\
= &- 2a^2(q^3 - q^2) + 2ab(q^2 - 1) - 2bc(q^2 - 1) + 2c^2(q + 1) \\
&+ p^2[-ab(q^2 + 1) - 2ac(q^2 - 1) + bc(q + 1)] \\
&+ pr[\ 6a^2(q^2 + q + 1) - ab(q - 1) + (14ac - 3b^2)(q + 1)] \\
&+ r^2\{-8a^2(q - 1) + 4ab\} \\
= &(q + 1)\{-2a^2(q^2 - q + q^2 - q + 1) + 2ab(q^2 - q + q - 1) - 2bc(q - 1) + 2c^2 \\
&+ p^2[-ab(q^2 - q - 1) - 2ac(q - 1) + bc] \\
&+ pr[\ 6a^2(q^2 + q + 1) - ab(q - 1) + (14ac - 3b^2)] \\
&+ r^2\{-8a^2(q - 1) + 4ab\}\}.
\end{aligned}$$

Hence writing, first, $a = c = 1$, $b = -\frac{1}{2}$, we obtain, after some reductions,

$$\Omega = (q + 1)\{-2q(q - 1)(q^2 - \tfrac{1}{4}q + 1) + p^2(q - 1)(-\tfrac{1}{2}q - 2) + pr(6q^2 - \tfrac{1}{4}q - 10) + r^2(-\tfrac{4}{3} + \tfrac{1}{2})\};$$

secondly, writing $a = c = 1$, $b = -2$, we obtain

$$\Omega = (q + 1)\{-2(q + 1)^2(q^2 + 1) + p^2 \cdot 2(q - 1)^2 + 2pr(3q^2 - 2q + 6) - 8r^2q\};$$

and, thirdly, writing $a = 1$, $b = -\frac{1}{2}$, $c = -\frac{1}{2}$, we obtain

$$\Omega = (q + 1)\{(-2q^3 + q^2 - \tfrac{3}{2}q + \tfrac{3}{2}q) + p^2(-\tfrac{1}{2}q^2 + \tfrac{3}{2}q - \tfrac{1}{2}q) + pr(6q^2 - \tfrac{15}{2}q - \tfrac{17}{4}) + r^2(-8q + \tfrac{5}{2})\}.$$

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652.

ON A SEXTIC TORSE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 229–235.]

(Page 68.) THE torse having for its edge of regression or cuspidal edge the curve defined by the equations $x = \cos\phi$, $y = \sin\phi$, $z = \cos 2\phi$, is an interesting and convenient one for the construction of a model, and it is here considered partly from that point of view.

The edge is a quadriquadric curve, the intersection of the cylinder $x^2+y^2 = 1$ with the parabolic hyperboloid $z = x^2 - y^2$; the cylinder regarded as a cone having its vertex at infinity on the line $x = 0$, $y = 0$, viz. the vertex is on the hyperboloid, or the curve is a nodal quadriquadric (the node being thus an isolated point at infinity on the line in question), and the torse is consequently of the order $8 - 2, = 6$, viz. it is a sextic torse.

The edge is a bent oval situate on the cylinder $x^2 + y^2 = 1$, such that, regarding ϕ as the azimuth (or angle measured along the circular base from its intersection with the axis of x), the altitude z is given by the equation $z = \cos 2\phi$; viz. there are in the plane xz, or, say in the planes xz, $x'z$, two maxima altitudes $z = 1$, and in the plane yz, or, say in the planes yz and $y'z$, two minima altitudes $z = -1$. The sections by these principal planes are, as is seen at once, nodal curves on the surface; they are, in fact, the cubic curves $z = 3 - \frac{2}{x^2}$, viz. here as x increases from ± 1 to $\pm\infty$, z increases from the before-mentioned value 1 to 3, and $z = -3 + \frac{2}{y^2}$, viz. as

y increases from ± 1 to $\pm\infty$, z decreases from the before-mentioned value -1 to -3. The two half-sheets (which meet in the cuspidal edge) intersect each other along these nodal lines, in such wise that the section of the surface by any axial plane (plane through the line $x = 0$, $y = 0$) is a curve having a cusp on the cuspidal edge, and such that when the axial plane coincides with either of the principal planes $x = 0$, $y = 0$, the

(Page 69.) two half-branches of the curve coincide together with the portions which lie outside the cylinder $x^2 + y^2 = 1$, in fact, the portions referred to above, of the nodal curve in the plane in question; the portions which lie inside the cylinder are acnodal or isolated curves without any real sheet through them. It may be added, in the way of general description, that the section of the surface by any cylinder $x^2 + y^2 = c^2$ $(c > 1)$ is a curve of the form $z = C\cos(2\theta \pm B)$, θ the angle along the base of the cylinder from the intersection with the axis of x; C, B are functions of c; viz. we have for the two half sheets respectively

$$z = C\cos(2\theta + B) \text{ and } z = C\cos(2\theta - B),$$

each curve having thus the two maxima $+C$, and the two minima $-C$; and the two curves intersect each other at the four points in the two principal planes respectively; viz. the points for which $\theta = 0, 90°, 180°, 270°$, and $z = C\cos B,\ -C\cos B,\ C\cos B,\ -C\cos B$ accordingly.

Proceeding to discuss the surface analytically, we have for the equations of a generating line

$$\frac{x - \cos\phi}{-\sin\phi} = \frac{y - \sin\phi}{\cos\phi} = \frac{z - \cos 2\phi}{-2\sin 2\phi} = \rho \text{ suppose,}$$

or say

$$\begin{aligned} x &= \cos\phi - \rho\sin\phi, \\ y &= \sin\phi + \rho\cos\phi, \\ z &= \cos 2\phi - 2\rho\sin 2\phi, \end{aligned}$$

which equations, considering therein ρ, ϕ as arbitrary parameters, determine the surface.

Writing $x = 0$, we find $y = \frac{1}{\cos\phi}$, and then $z = -3 + 2\sin^2\phi$, viz. we have

$$x = 0,\ z = -3 + \frac{2}{y^2}, \text{ for section in plane } yz,$$

and, similarly, writing $y = 0$, we find $x = \frac{1}{\cos\phi}$, and then $z = 3 - 2\cos^2\phi$, viz.

$$y = 0,\ z = 3 - \frac{2}{x^2}, \text{ for section by plane } xz.$$

By what precedes, these are nodal curves, crunodal for the portions

$$(y = \pm 1 \text{ to } \pm\infty,\ z = -1 \text{ to } -3) \text{ and } (x = \pm 1 \text{ to } \pm\infty,\ z = 1 \text{ to } 3)$$

respectively, acnodal for the remaining portions $y < \pm 1$, $x < \pm 1$ respectively.

Writing $x = r\cos\theta$, $y = r\sin\theta$, so that the coordinates of a point on the surface are r, θ, z, where $r = \sqrt{(x^2 + y^2)}$ is the projected distance, θ is the azimuth from the axis of x, and z is the altitude, we have

$$\begin{aligned} r\cos\theta &= \cos\phi - \rho\sin\phi, \\ r\sin\theta &= \sin\phi + \rho\cos\phi, \\ z &= \cos 2\phi - 2\rho\sin 2\phi. \end{aligned}$$

(Page 70.) We have $r^2 = 1 + \rho^2$; and thence also, if $\tan\alpha = 2\rho, = \pm 2\sqrt{(r^2-1)}$, that is,

$$\cos\alpha = \frac{1}{\sqrt{(4r^2-3)}}, \quad \sin\alpha = \pm\frac{2\sqrt{(r^2-1)}}{\sqrt{(4r^2-3)}},$$

then

$$z = \sqrt{(4r^2-3)}\cos(2\theta+\beta),$$

showing that for a given value of r (or section by the cylinder $x^2+y^2=r^2$) the maximum and minimum values of z are $z = \pm\sqrt{(4r^2-3)}$.

But proceeding to eliminate ϕ, we find

$$r^2\cos 2\theta = (1-\rho^2)\cos 2\phi - 2\rho\sin 2\phi,$$
$$r^2\sin 2\theta = 2\rho\cos 2\phi + (1-\rho^2)\sin 2\phi$$

or multiplying these by $1+3\rho^2$ and 2ρ and adding

$$r^2\{(1+3\rho^2)\cos 2\theta + 2\rho^2\sin 2\theta\} = (1+\rho^2)^2(\cos 2\phi - 2\rho\sin 2\phi),$$

that is,

$$r^2\{(3r^2-2)\cos 2\theta \pm 2(r^2-1)^{\frac{1}{2}}\sin 2\theta\} = r^4 z;$$

or, finally,

$$r^2 z = (3r^2-2)\cos 2\theta + 2(r^2-1)^{\frac{1}{2}}\sin 2\theta,$$

which is the equation of the surface in terms of the coordinates r, θ, z.

Observing that $(3r^2-2)^2 + 4(r^2-1)^3 = r^4(4r^2-3)$, we may write

$$r^2\sqrt{(4r^2-3)}\cos\beta = 3r^2-2,$$
$$r^2\sqrt{(4r^2-3)}\sin\beta = 2(r^2-1)^{\frac{3}{2}},$$

and therefore also

$$\tan\beta = \frac{2(r^2-1)^{\frac{3}{2}}}{3r^2-2},$$

and the equation thus becomes

$$z = \sqrt{(4r^2-3)}\cos(2\theta+\beta),$$

where z is the altitude belonging to the azimuth θ in the cylindrical section, radius r. The maxima and minima altitudes are $\pm\sqrt{(4r^2-3)}$, and these correspond to the values $\theta = \pm\frac{1}{2}\beta$, $\frac{1}{2}\pi + \frac{1}{2}\beta$, $\pi + \frac{1}{2}\beta$, $\frac{3}{2}\pi + \frac{1}{2}\beta$; it is to be further noticed that when $r = 1$, we have $\beta = 0$, but as r increases and becomes ultimately infinite, β increases to $\frac{1}{2}\pi$, that is, $\frac{1}{2}\beta$ increases from 0 to $\frac{1}{4}\pi$.

It may be noticed that the surface is a peculiar kind of deformation, obtained by giving proper rotations to the several cylindrical sections of the surface $z = \sqrt{(4r^2-3)}\cos 2\theta$; viz. in rectangular coordinates this is $r^2 z = \sqrt{(4r^2-3)}(x^2-y^2)$, that is,

$$(x^2+y^2)^4 z^2 - \{4(x^2+y^2)-3\}(x^2-y^2)^2 = 0.$$

To obtain the equation in rectangular coordinates, we have

$$r^4 z^2 - 2z(3r^2-2)(x^2-y^2) + (3r^2-2)^2 - 16(r^2-1)^3\sin^2 2\theta = 0,$$

viz. this is

$$r^4 z^2 - 2z(3r^2-2)(x^2-y^2) + (3r^2-2)^2\left(1 - \frac{4x^2y^2}{r^4}\right) - 16(r^2-1)^3\frac{4x^2y^2}{r^4} = 0,$$

(Page 71.) or, what is the same thing, it is

$$r^4z^2 - 2z(3r^2-2)(x^2-y^2) + (3r^2-2)^2 - \frac{4x^2y^2}{r^4}\{4(r^2-1)^3 + (3r^2-2)^2\} = 0,$$

viz. the term in { } being $r^4(4r^2-3)$, this is

$$r^4z^2 - 2z(3r^2-2)(x^2-y^2) + (3r^2-2)^2 - 4x^2y^2(4r^2-3) = 0,$$

or say

$$z^2(x^2+y^2)^2 - 2z(3x^2+3y^2-2)(x^2-y^2) + (3x^2+3y^2-2)^2 - 4x^2y^2(4x^2+4y^2-3) = 0.$$

This may also be written

$$\{z(x^2-y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2y^2(z^2 - 4x^2 - 4y^2 + 3) = 0,$$

a form which puts in evidence the nodal curves

$$x = 0,\ zy^2 = -3y^2 + 2,\ \text{and } y = 0,\ zx^2 = 3x^2 - 2.$$

It shows also that the quadric cone $z^2 - 4x^2 - 4y^2 + 3 = 0$ touches the surface along the curve of intersection with the surface $z(x^2-y^2) - 3(x^2+y^2) + 2 = 0$. This is, in fact, the curve of maxima and minima of the cylindrical sections, viz. reverting to the form $z = \sqrt{(4r^2-3)}\cos(2\theta+\beta)$, or, if for greater clearness, attending only to one sheet of the surface, we write it $z = \sqrt{(4r^2-3)}\cos(2\theta-\beta)$, we have a maximum, $z = \sqrt{(4r^2-3)}$, for $2\theta = \beta$ (or $2\pi+\beta$), giving

$$\cos 2\theta = \cos\beta = \frac{3r^2-2}{r^2\sqrt{(4r^2-3)}} = \frac{3r^2-2}{r^2z},$$

and a minimum, $z = -\sqrt{(4r^2-3)}$, for $2\theta = \pi+\beta$ (or $3\pi+\beta$), giving

$$\cos 2\theta = -\cos\beta = -\frac{3r^2-2}{r^2\sqrt{(4r^2-3)}} = \frac{3r^2-2}{r^2z},$$

viz. the locus is $z^2 = 4(r^2-3)$, $z(x^2-y^2) = 3r^2-2$; and for $z = \sqrt{(4r^2-3)}\cos(2\theta+\beta)$ we find the same locus, viz. the equations of the locus are

$$z^2 - 4x^2 - 4y^2 + 3 = 0, \quad z(x^2-y^2) - 3x^2 - 3y^2 + 2 = 0,$$

as above.

To put in evidence the cuspidal edge, write for a moment $\zeta = z - x^2 + y^2$, the equation becomes

$$\{\zeta(x^2-y^2) + (r^2-1)(r^2-2) - 4x^2y^2\}^2 + 4x^2y^2\{\zeta^2 + 2\zeta(x^2-y^2) + (r^2-1)(r^2-3) - 4x^2y^2\} = 0;$$

viz. this is

$$\zeta^2r^4 + 2\zeta(x^2-y^2)(r^2-1)(r^2-2) + (r^2-1)^2(r^2-2)^2 - 4\zeta^2x^2y^2(r^2-1)^2 = 0,$$

or writing the last term thereof in the form

$$-[r^4 - (x^2-y^2)^2](r^2-1)^2,$$

and then putting $r^2 = 1 + U$, the equation is

$$\zeta^2(1+2U+U^2) + 2\zeta U(U-1)(x^2-y^2) + U^2(U-1)^2 - U^2\{(U+1)^2 - (x^2-y^2)^2\} = 0;$$

viz. this is

$$\{\zeta - U(x^2 - y^2)\}^2 + 2U\{\zeta^2 + \zeta U(x^2 - y^2) - 2U^2\} + 4U^2 = 0,$$

(Page 72.) showing the cuspidal edge $\zeta = 0, U = 0$, viz. $z = x^2 - y^2$, $x^2 + y^2 = 1$. Moreover, along the cuspidal edge the surface is touched by $\zeta - U(x^2 - y^2) = 0$, that is, by $z - (x^2 - y^2) = 0$; and at the points where this tangent surface again meets the surface we have $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$; viz. the surface contains upon itself the curve represented by this last equation, and $z - (x^2 - y^2) = 0$.

As a verification, in the form

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2y^2(z^2 - 4x^2 - 4y^2 + 3) = 0$$

of the equation of the surface, write $z = x^2 - y^2$. If for a moment $x^2 + y^2 = \lambda$, $x^2 - y^2 = \mu$, then the value of z is $z = \lambda\mu$, and the equation becomes

$$(\lambda\mu^2 - 3\lambda + 2)^2 + (\lambda^2 - \mu^2)(\lambda^2\mu^2 - 4\lambda + 3) = 0,$$

that is,

$$\mu^2(\lambda^4 - 6\lambda^2 + 8\lambda - 3) - 4\lambda^3 + 12\lambda^2 - 12\lambda + 4 = 0;$$

or, what is the same thing,

$$(\lambda - 1)^2\{\mu^2(\lambda + 3) - 4\} = 0,$$

so that we have $(\lambda - 1)^2 = 0$, or else $\mu^2(\lambda + 3) - 4 = 0$; viz. $(x^2 + y^2 - 1)^2 = 0$, or else $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$, agreeing with the former result.

In polar coordinates, the surface is touched along the cuspidal edge by the surface $z = r^2 \cos 2\theta$, and where this again meets the surface we have $r^4(r^2 + 3)\cos^2 2\theta - 4 = 0$.

For the model, taking the unit to be 1 inch, I suppose that for the edge of regression we have

$$x = 2\cos\phi, \quad y = 2\sin\phi, \quad z = 5 + (\cdot 45)\cos 2\phi,$$

viz. the curve is situate on a cylinder radius 2 inches. And I construct in zinc-plate the cylindric sections, or say the templets, for one sheet of the surface, for the several radii $2, 3, \ldots, 8$ inches; taking the radius as k inches, the circumference of the cylinder, or entire base of the flattened templet, is $= 2k\pi$; and the altitude, writing 2θ in place of $2\theta - \beta$ as above, is given by the formula $z = 5 + (\cdot 45)\sqrt{(k^2 - 3)}\cos 2\theta$, so that the half altitude of the wave is $= (\cdot 45)\sqrt{(k^2 - 3)}$; having this value, the curve is at once constructed geometrically. We have, moreover, $\cos\beta = \frac{3k^2 - 8}{k^2\sqrt{(k^2 - 3)}}$; the numerical values then are

k	$2k\pi$	$(\cdot 45)\sqrt{(k^2 - 3)}$	$\dfrac{3k^2 - 8}{k^2\sqrt{(k^2 - 3)}}$	$\frac{1}{2}\beta$
2	12.57	0.45	1.00	0°
3	18.85	1.10	.86	15
4	25.13	1.62	.69	23
5	31.42	2.11	.57	$27\frac{1}{2}$
6	37.70	2.59	.48	$30\frac{1}{2}$
7	43.98	3.05	.42	$32\frac{1}{2}$
8	50.27	3.51	.36	34

the altitudes in the successive templets being thus included between the limits $5 \pm 0 \cdot 45$, $5 \pm 1 \cdot 10, \ldots, 5 \pm 3 \cdot 51$.

653.

ON A TORSE DEPENDING ON THE ELLIPTIC FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 235–241.]

(Page 73.) ON attempting to cover with paper one half-sheet of the foregoing sextic torse, [652], I found that the paper assumed approximately the form of a circular annulus of an angle exceeding 360°, and this led me to consider the general theory of the construction of a torse in paper, and, in particular, to consider the torses such that when developed into a plane the edge of regression becomes a circular arc. It is scarcely necessary to remark that, to construct in paper a circular annulus of an angle exceeding 360°, we have only to take a complete annulus, cut it along a radius, and then insert (gumming it on to the two terminal radii) a portion of an equal circular annulus; drawing from each point of the inner circular boundary a half-tangent, and considering these half-tangents as rigid lines, the paper will bend round them so as to form the half-sheet of a torse having for its edge of regression this inner boundary, which will assume the form of a closed curve with two equal and opposite maxima and two equal and opposite minima, described on a cylinder, and being approximately such as the curve given by the equations

$$x = \cos\theta, \quad y = \sin\theta, \quad z = m\cos 2\theta.$$

Considering, in general, an arc PQ (without inflexions) of any curve, and drawing at the consecutive points P, P', P'', &c. the several half-tangents PT, $P'T'$, $P''T''$, …, then, considering these as rigid lines and bending the paper round them, we have the half-sheet of a torse, having for its edge of regression the curve in question now bent into a curve of double curvature. It is, moreover, clear that the edge of regression has at each point thereof the same radius of absolute curvature as the original plane curve; in fact, if in the plane curve $PP' = ds$, and the angle $T'PT$ between the consecutive half-tangents PT and $P'T'$ be $= d\phi$, these quantities ds and

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(Page 74.) $d\phi$ remain unaltered in the curve of double curvature; and the radius of absolute curvature is given by the equation $\rho\, d\phi = ds$. In particular when, as above, the arc is a circular one, say of radius $= a$, then, however the paper is bent, the edge of regression has at each point thereof the radius of absolute curvature $= a$.

Consider on any given surface, at a given point P thereof, and in a given direction, an element of length PP', then (under the restrictions presently mentioned) we can determine the consecutive element $P'P''$, such that the curve $PP'P''\ldots$ shall have at P a radius of absolute curvature $= a$; in fact, r being the radius of curvature of the normal section of the surface through the element PP', the radius of curvature of the section inclined at an angle θ to the normal section is $= r\cos\theta$; so that we have only to take the section at the inclination $\theta, = \cos^{-1}\frac{a}{r}$ to the normal section, and we have the consecutive element $P'P''$ such that the radius of absolute curvature of the curve $PP'P''$ is $= a$. The necessary restriction, of course, is that $r > a$; thus, if at the given point P the two principal radii of curvature are of the same sign (to fix the ideas, let the two principal radii and also a be each of them positive), then we may on the surface determine a direction PQ, for which the radius of curvature of the normal section is $= a$; and then the direction of the element PP' may be any direction between PQ and the direction PR, corresponding to the greatest of the two principal radii.

Having obtained the element $P'P''$, we may, if the radius of absolute curvature at P' be given, construct the next element $P'P'''$, and so on; that is to say, on a given surface starting from a given point P and given initial direction PP', we can (under a restriction, as above, as to the curvature at the different points of the surface) construct a curve having at the successive points thereof given values of the radius of absolute curvature; viz., the value may be given either as a function of the coordinates of the point on the surface, or as a function of the length of the curve measured say from the initial point P; it is in this last manner that in what follows the value of the radius of absolute curvature is assumed to be given.

We may thus, taking on paper an arc PQ with its half-tangents, apply it to a given surface, the point P to a given point, and the infinitesimal arc PP' to an element PP' in a given direction from the given point; and we thus obtain the half-sheet of a torse having for its edge of regression a determinate curve upon the surface. In particular, the arc PQ may be circular of the radius a, and the surface be a circular cylinder of radius α; and we thus obtain the torse having for edge of regression a curve on the cylinder radius α, and such that the radius of absolute curvature is at each point $= a$. There are three cases according as $\alpha > a$, $\alpha = a$, or $\alpha < a$; it is to be remarked that if $\alpha > a$, then the curve must at each point cut the generating line of the cylinder at an angle not exceeding $\cos^{-1}\frac{a}{\alpha}$, but that in the other two cases the angle may have any value whatever; and, further, that in every case when the angle is $= 0$, viz. when the curve touches a generating line of the cylinder, then the osculating plane of the curve coincides with the tangent plane of the cylinder.

(Page 75.) The analytical theory is very simple. Taking x, y, z as functions of the length s, we have

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1;$$

the condition, which expresses that the radius of absolute curvature is $= a$, then is

$$\left(\frac{d^2x}{ds^2}\right)^2 + \left(\frac{d^2y}{ds^2}\right)^2 + \left(\frac{d^2z}{ds^2}\right)^2 = \frac{1}{a^2}.$$

By what precedes, the point (x, y, z) may be taken to be upon a given surface, say upon the cylinder $x^2 + y^2 = \alpha^2$; we may then write $x = \alpha\cos\theta$, $y = \alpha\sin\theta$. Taking instead of s any independent variable u whatever, and using accents to denote the derived functions in regard to u, the equations become

$$\begin{aligned} x'^2 + y'^2 + z'^2 &= s'^2, \\ x''^2 + y''^2 + z''^2 - s''^2 &= \frac{1}{a^2}s'^4, \\ &\qquad x = \alpha\cos\theta, \quad y = \alpha\sin\theta. \end{aligned}$$

From the last two equations we obtain

$$x'^2 + y'^2 = \alpha^2\theta'^2, \quad x''^2 + y''^2 = \alpha^2(\theta''^2 + \theta'^4),$$

and the first two equations thus become

$$\alpha^2\theta'^2 + z'^2 = s'^2,$$

$$\alpha^2(\theta''^2 + \theta'^4) + z''^2 - s''^2 = \frac{1}{a^2}s'^4,$$

and from the first of these we find

$$s'' = \frac{\alpha^2\theta'\theta'' + z'z''}{(\alpha^2\theta'^2 + z'^2)^{\frac{1}{2}}},$$

whence the second equation is

$$\alpha^2(\theta''^2 + \theta'^4) + z''^2 - \frac{(\alpha^2\theta'\theta'' + z'z'')^2}{\alpha^2\theta'^2 + z'^2} = \frac{1}{a^2}(\alpha^2\theta'^2 + z'^2)^2,$$

or reducing, this is

$$(\alpha^2\theta'^2 + z'^2)(\theta''^2 + \theta'^4) + (z''^2\theta'^2 - 2\theta'\theta''z'z'' - \alpha^2\theta'^2\theta''^2) = \frac{1}{a^2}(\alpha^2\theta'^2 + z'^2)^2.$$

Taking θ as the independent variable, we have $\theta' = 1$, $\theta'' = 0$, and the equation becomes

$$\alpha^2 + z'^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^2,$$

or, what is the same thing,

$$z''^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^3 - (\alpha^2 + z'^2)^2.$$

Write here

$$\alpha^2 + z'^2 = \Omega^2,$$

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